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Due Date: 11 December 2024
Lab Section: 03

Final Assignment: Reintroduction of the Razorback Quahog in the Plum Island Estuary

Introduction:

The purpose of this assignment was to help the Plum Island Shellfish Cooperative (PISCO) to re-introduce populations of shellfish called *razorback quahog* which got decimated back in 2002. We had to help PISCO scope out potential sites out of a total 20, which would be considered safe for the introduction of quahog in the Palm Island Estuary in Northeastern Massachusetts. This analysis considers ecological and hydrological factors to determine the most viable sites among 20 potential locations. There are four major criteria that the study considers in order to identify the potential safe sites. These include: salt marsh evaluation around potential sites, stormwater discharge impacts, eutrophication risk assessment, and the proximity of sites to Long Term Ecological Research (LTER) stations.

Methods:

Salt Marsh Evaluation: Using the pairwise buffer tool, we identified **14 out of the 20** potential sites that met the minimum requirement of 5 hectares of salt marsh area, as detailed in *Table 1* and also shown in *map 1*. These sites provide suitable habitat for the saltmarsh stickleback fish, which is believed to play a crucial role in mitigating pathogen transmission—the primary cause of the previous quahog population collapse in the estuary. *Table 2* highlights the specific salt marsh area for each site, highlighting that only sites with an area of 5 hectares or more are considered safe.

Stormwater Discharge Impact: We used the distance accumulation tool to analyze how stormwater energy dissipates from the stream outlets. This tool helped calculate the cumulative resistance stormwater encounters as it spreads across the estuary. Each of the three gauge stations showed different dissipation distances, with energy dropping off at 748 meters, 1250 meters, and 450 meters, as shown in *Map 2.* Taking the maximum dissipation distance of 1250 meters into account from the combined stormwater impact from all three gauge stations, we found that **5** of the potential sites are located beyond the high-impact zones, making them safe from significant stormwater effects as displayed in *map 3* and *table 1.*

Eutrophication Risk Assessment: To assess the risk of eutrophication, we analyzed the ratio of nitrogen (N) to phosphorus (P) concentrations in the estuary, as these influence the likelihood of harmful algal blooms. Algal blooms can deplete oxygen levels when they die, which poses a threat to the reintroduced quahog population. First, we used the IDW (Inverse Distance Weighted) interpolation tool to estimate nitrogen and phosphorus concentrations in the estuary. These concentration maps can be seen in *Map 4*. Next, we calculated the N:P ratio for the area based on the interpolated data. Areas with an N:P ratio between **13:1 and 18:1** were classified as unsafe because these conditions are most likely to lead to algal blooms. As seen in *Map 5,* out of the 20 potential reintroduction sites, ***8 were located in these unsafe zones***. The other **12 sites** were in areas where the N:P ratio was outside this unsafe range, making them safer for quahog reintroduction. These findings are summarized in Table 1, which lists the sites safe for quahog reintroduction, and *Table 2,* which provides the specific N:P ratio for each site.

Proximity to LTER Stations: We used the **Near Tool**, which calculates the shortest distance from each potential site to the closest LTER site. This analysis provided precise measurements of proximity between the proposed reintroduction sites and the LTER stations. Out of the 20 proposed sites, 3 were deemed acceptable based on all the evaluation criteria. For these sites (Site IDs: 9, 18, and 20), we calculated the distances to their nearest LTER stations. These results are summarized in *Table 3*, which shows the distance of each acceptable site to its respective LTER station.

Results:

The multi-criteria analysis identified three optimal sites, with ***Site IDs 9, 18, and 20***, as shown in *Table 1,* for the reintroduction of razorback quahogs in the Plum Island Estuary. These sites were selected based on their sufficient salt marsh habitat, minimal stormwater impact, low risk of eutrophication, and close proximity to LTER stations, ensuring effective monitoring and management. The distribution of these safe and unsafe sites in the Palm Island Estuary is also illustrated in *Map 6.*

Potential Quahog Reintroduction sites in the Plum Island Estuary				
Sites ID	**Final Safe Quahog reintroduction Sites**	**Sites with 5 ha Salt Marsh around them** ***(Safe sites)***	**Sites falling in areas with NO Eutrophication** ***(Safe sites)***	**Stormwater criterion** ***(Safe sites)***
1			✓	
2				
3		✓	✓	
4		✓		
5		✓	✓	
6		✓		
7			✓	
8			✓	
9	✓	✓	✓	✓
10		✓	✓	
11			✓	
12				✓
13		✓		✓
14		✓	✓	
15		✓	✓	
16		✓		
17		✓		
18	✓	✓	✓	✓
19		✓		
20	✓	✓	✓	✓

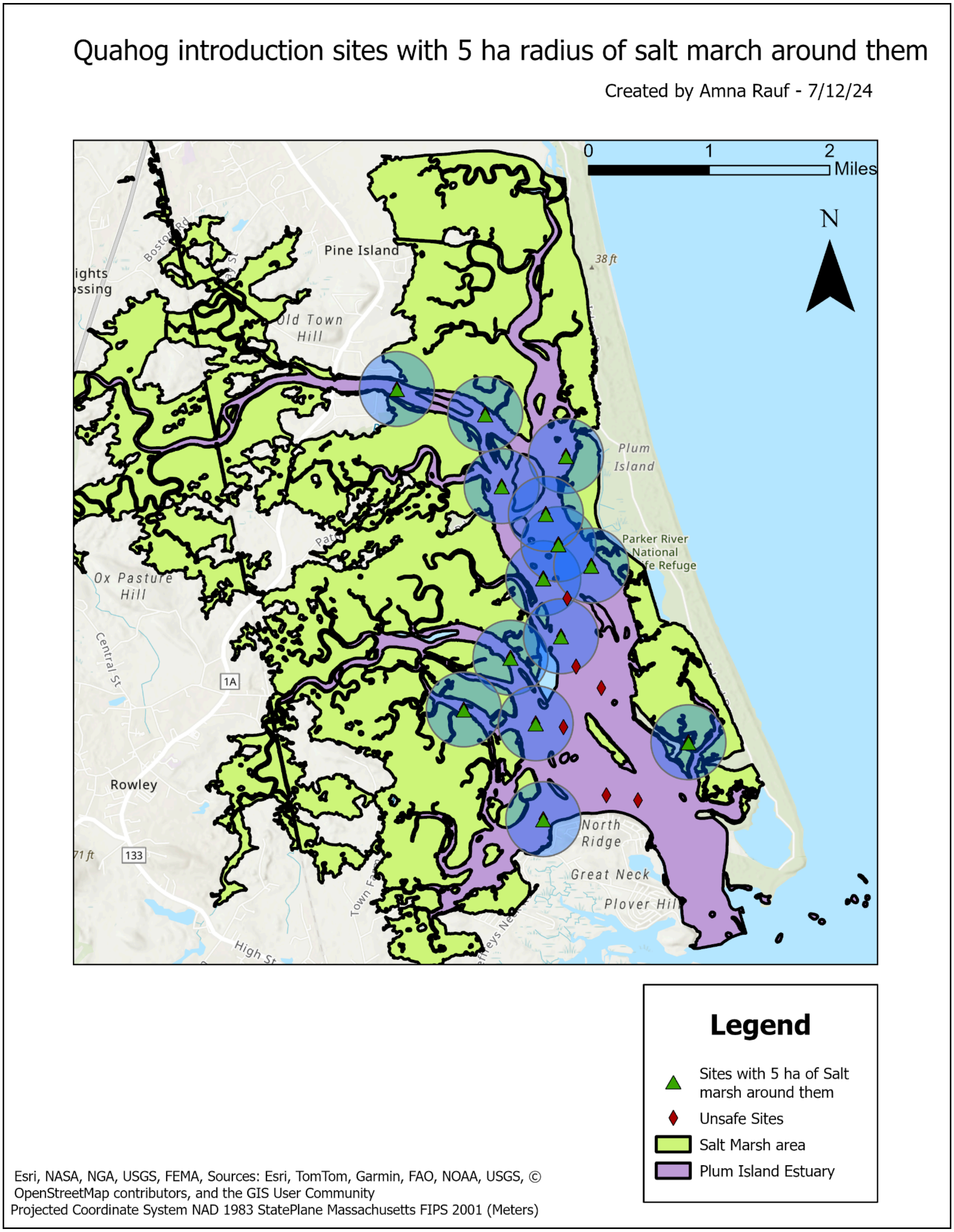
Table 1: Summary of Potential Quahog Reintroduction Sites and Their Suitability Based on Evaluation Criteria

Characteristics of Potential Quahog Sites		
Sites ID	**Salt Marsh (Hectares)**	**N/P Ratio of each site (uM/L)**
1	0.23	12.01
2	3.63	14.63
3	12.10	33.43
4	42.88	14.71
5	57.27	27.41
6	10.66	13.89
7	3.68	12.33
8	3.69	10.36
9	**28.51**	**39.58**
10	43.72	19.37
11	0.60	19.97
12	0.00	15.36
13	18.67	20.31
14	32.94	23.11
15	7.99	20.38
16	17.03	13.94
17	9.86	15.66
18	**46.03**	**7.91**
19	10.90	12.78
20	**11.50**	**20.05**

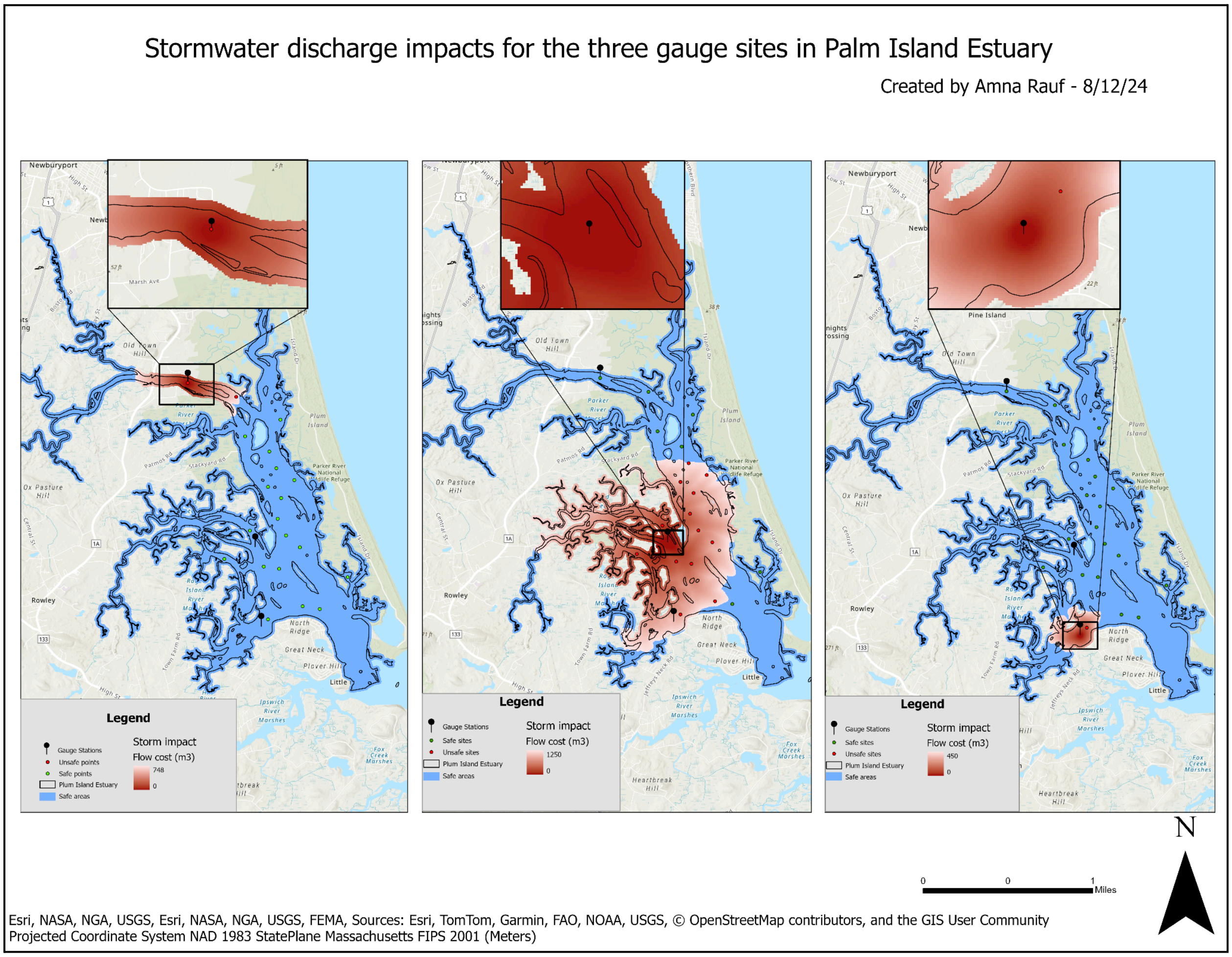
Table 2: Key Characteristics of Potential Quahog Reintroduction Sites: Salt Marsh Area and N:P Ratios

LTER sites closest to the acceptable quahog reintroduction sites				
Sites ID	**Acceptable quahog reintroduction sites**	**Station name**	**LTER Type**	**Distance (meters)**
9	✓	HTL-SO-5(Sub Headquarters)	Seine Sampling Stations	312.40
18	✓	EST-SO-30 IBYC	Estuary Sampling Station	1095.76
20	✓	HTL-SO-13	Fish Trawl Site	530.22

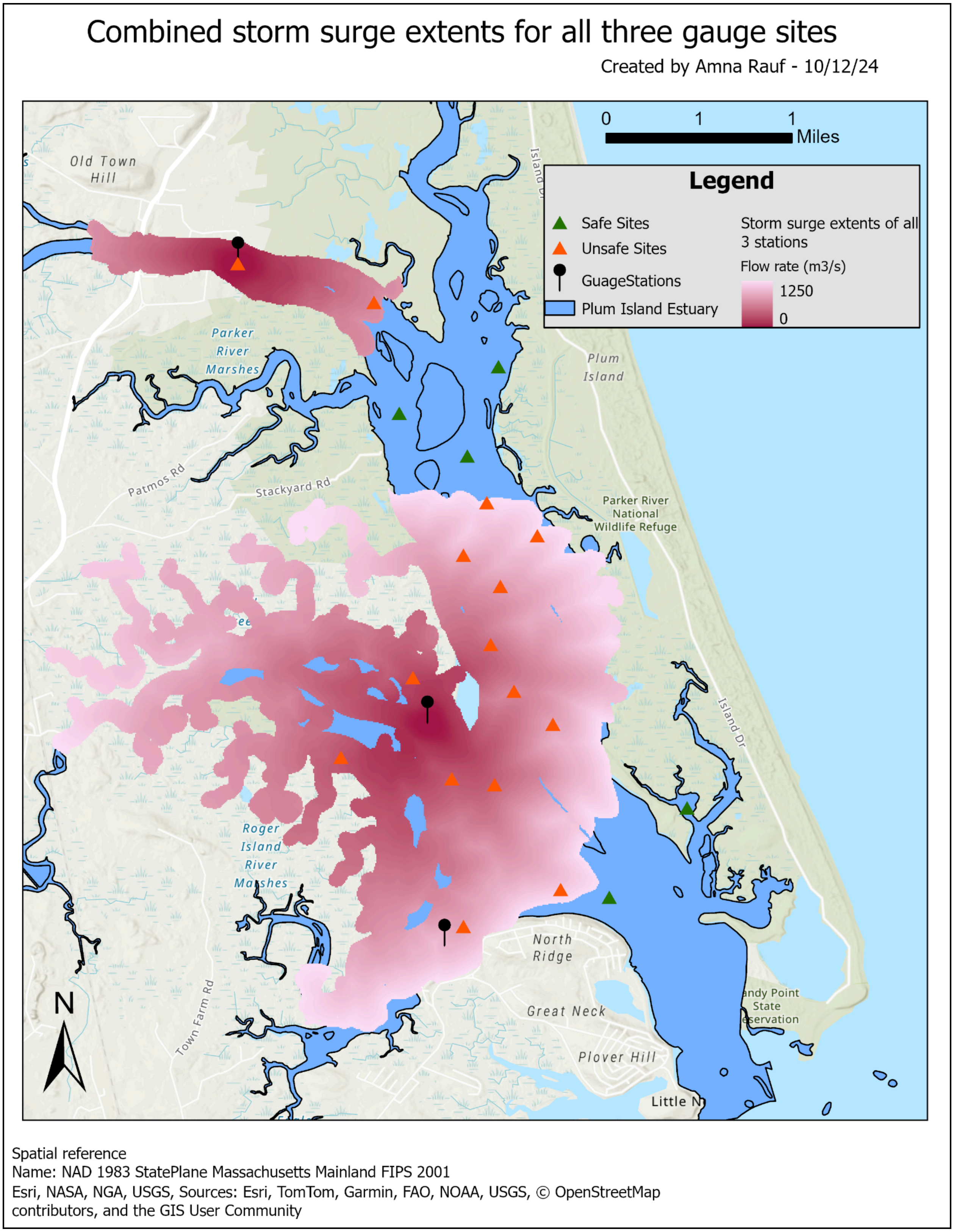
Table 3: Nearest LTER Sites to Acceptable Quahog Reintroduction Locations: Station names and Distances



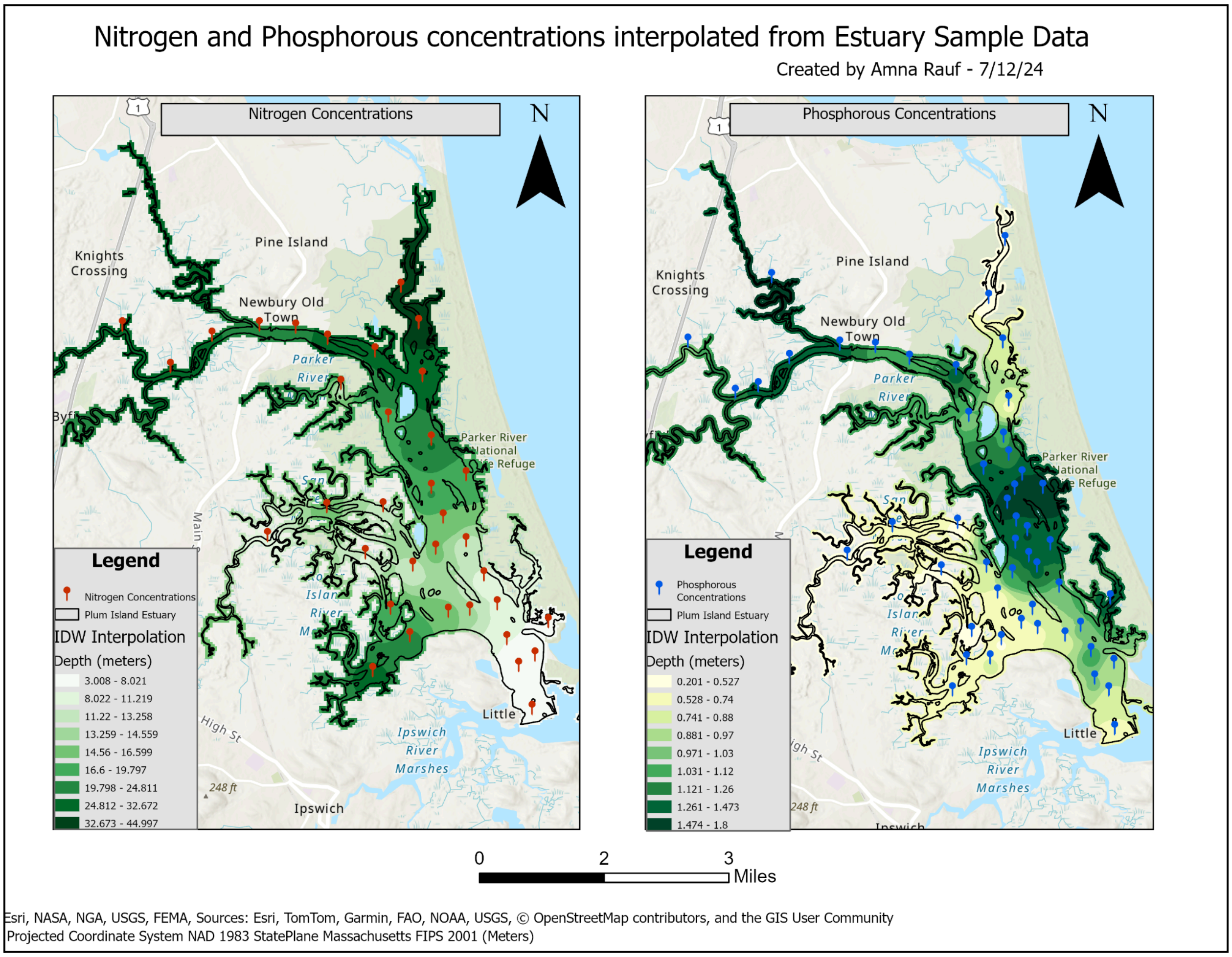
Map 1: Salt Marsh Suitability for Potential Quahog Reintroduction Sites in the Plum Island Estuary



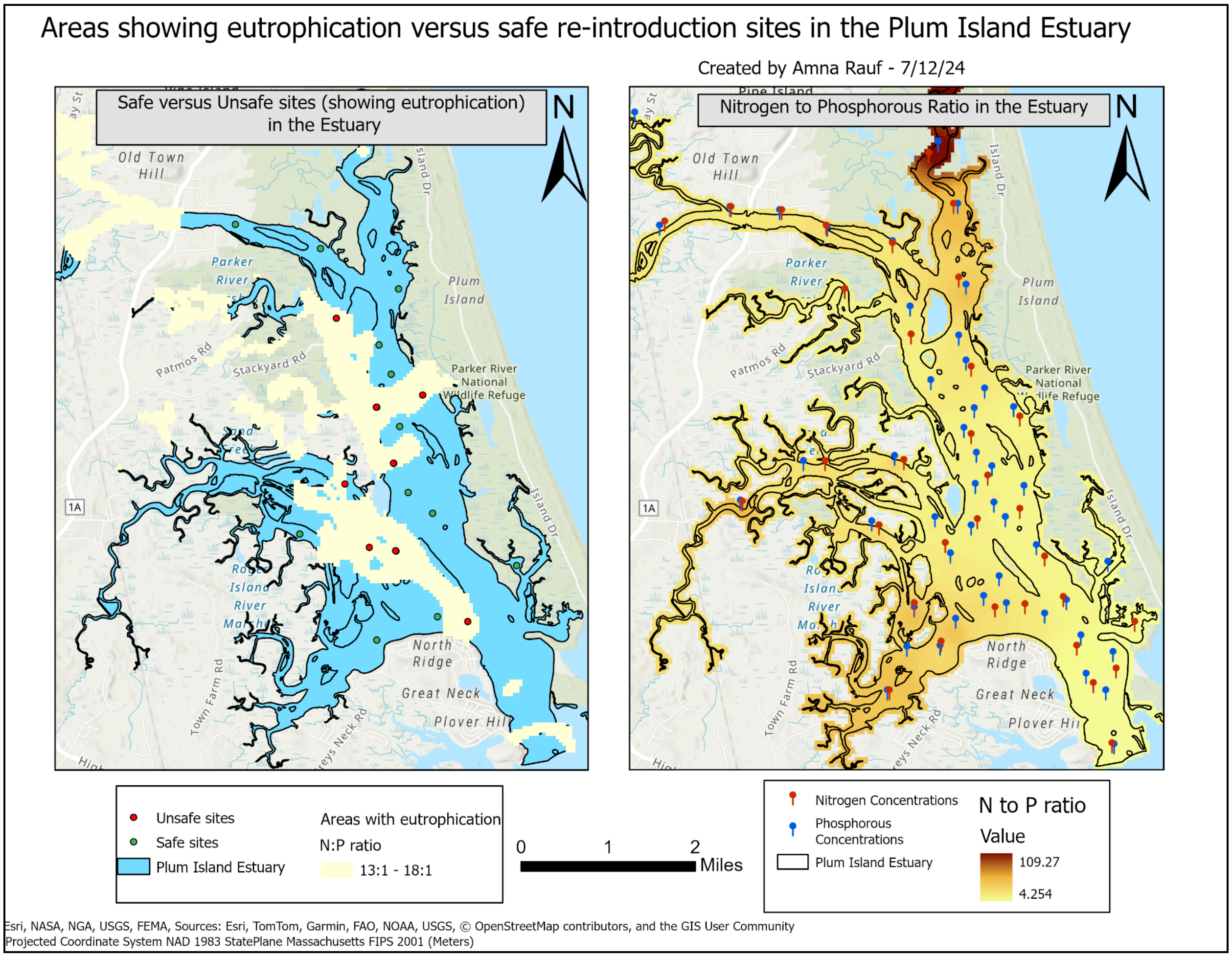
Map 2: Stormwater Discharge Impacts from Three Gauge Stations in the Plum Island Estuary



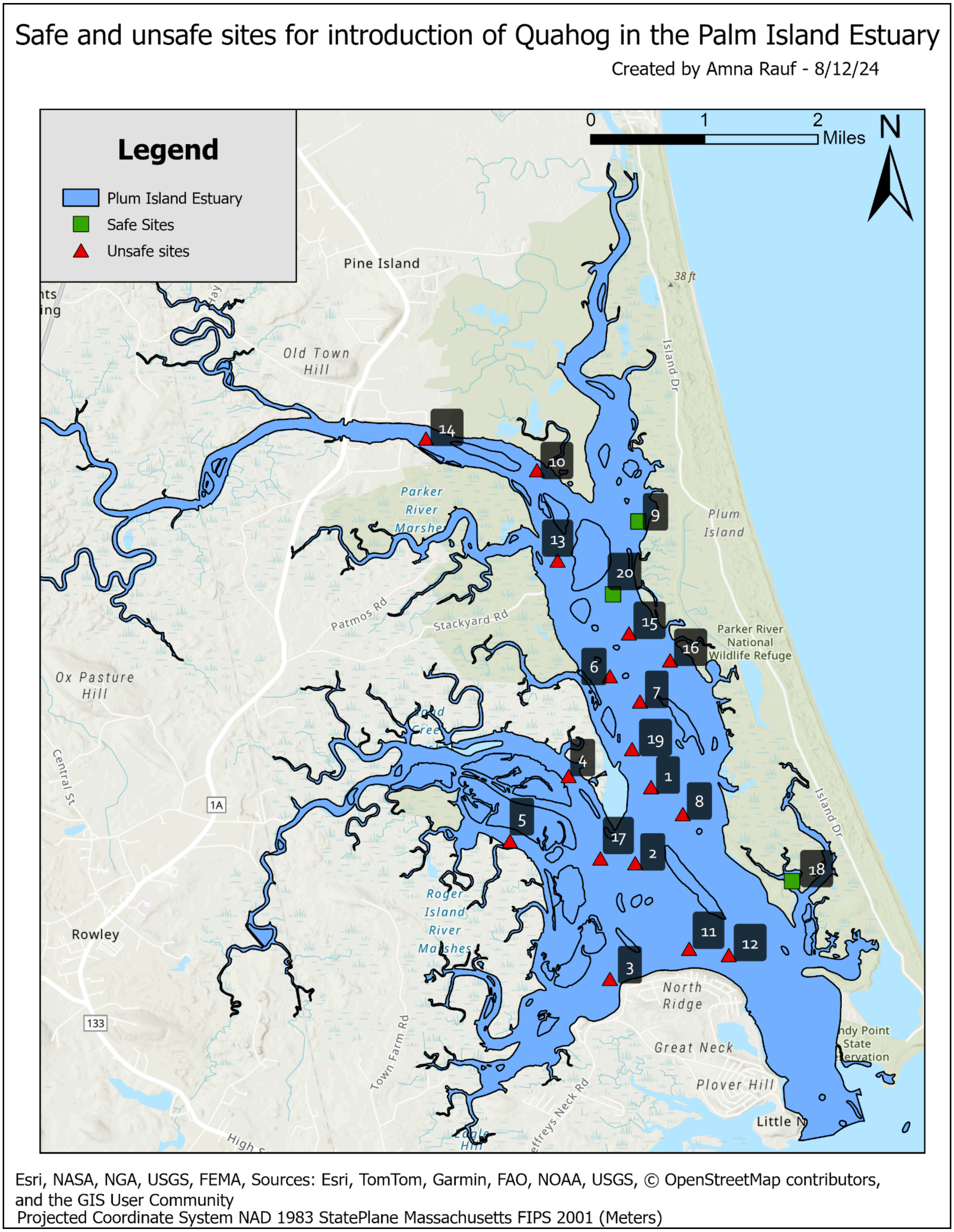
Map 3: Combined Stormwater impact from all three gauge stations and Suitability of Potential Quahog Reintroduction Sites



Map 4:Nitrogen and Phosphorus Concentrations Interpolated Across the Plum Island Estuary

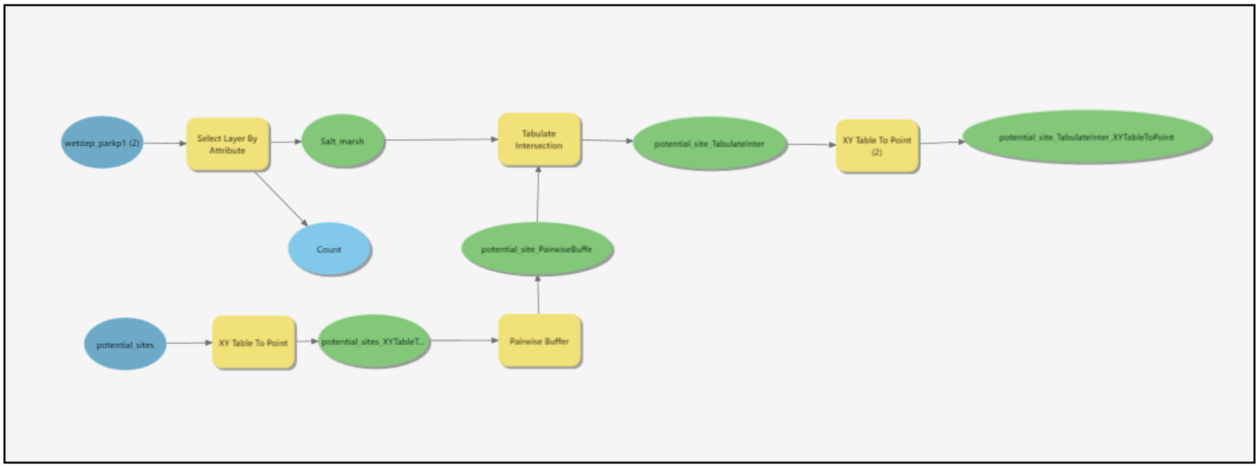


Map 5: N:P Ratio and Eutrophication Risk for Potential Quahog Reintroduction Sites



Map 6: Safe and Unsafe Quahog Reintroduction Sites in the Plum Island Estuary

Appendix:



Model 1: Using the pairwise buffer tool to identify sites with 5 hectares of salt marsh area

Model Script:

Please find the python script that defines the model displayed in Model 1:

```
# -*- coding: utf-8 -*-
"""
Generated by ArcGIS ModelBuilder on : 2024-12-07 17:57:20
"""
import arcpy


def Task1():  # Task1


    # To allow overwriting outputs change overwriteOutput option to True.
    arcpy.env.overwriteOutput = False
```

```
    arcpy.ImportToolbox(r"c:\program files\arcgis\pro\Resources\ArcToolbox\toolboxes\Data Management Tools.tbx")

    # Model Environment settings

    with arcpy.EnvManager(extent="246765.265388443 938377.187278361 260557.296761288 949658.875218475 PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532925199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAMETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]", outputCoordinateSystem="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532925199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAMETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]"):

        wetdep_parkp1_2_ = "wetdep_parkp1"

        potential_sites = "potential_sites"


        # Process: Select Layer By Attribute (Select Layer By Attribute) (management)

        Salt_marsh, Count = arcpy.management.SelectLayerByAttribute(in_layer_or_view=wetdep_parkp1_2_, where_clause="IT_VALDESC = 'SALT MARSH'")


        # Process: XY Table To Point (XY Table To Point) (management)

        potential_sites_XYTableToPoint = "P:\\Final_Nov_24\\Final\\Final.gdb\\potential_sites_XYTableToPoint"

        arcpy.management.XYTableToPoint(in_table=potential_sites, out_feature_class=potential_sites_XYTableToPoint, x_field="STATE_X", y_field="STATE_Y", coordinate_system="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532
```

925199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAMETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]];-36530900 -28803200 122610652.811951;-100000 10000;-100000 10000;0.001;0.001;0.001;IsHighPrecision")

 # Process: Pairwise Buffer (Pairwise Buffer) (analysis)

potential_site_PairwiseBuffe = "P:\\Final_Nov_24\\Final\\Final.gdb\\potential_site_PairwiseBuffe"

arcpy.analysis.PairwiseBuffer(in_features=potential_sites_XYTableToPoint, out_feature_class=potential_site_PairwiseBuffe, buffer_distance_or_field="500 Meters")

 # Process: Tabulate Intersection (Tabulate Intersection) (analysis)

potential_site_TabulateInter = "P:\\Final_Nov_24\\Final\\Final.gdb\\potential_site_TabulateInter"

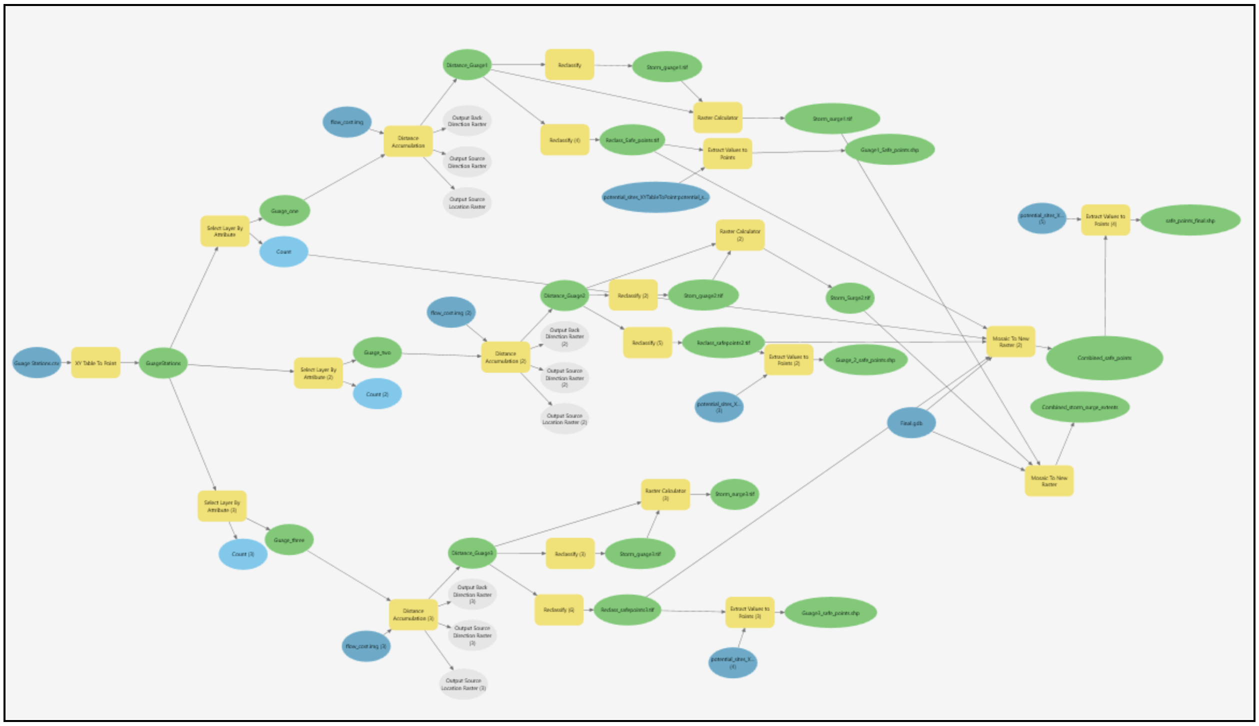
arcpy.analysis.TabulateIntersection(in_zone_features=potential_site_PairwiseBuffe, zone_fields=["Sites_ID"], in_class_features=Salt_marsh, out_table=potential_site_TabulateInter, sum_fields=["AREA_ACRES"], out_units="HECTARES")

 # Process: XY Table To Point (2) (XY Table To Point) (management)

potential_site_TabulateInter_XYTableToPoint = "P:\\Final_Nov_24\\Final\\Final.gdb\\potential_site_TabulateInter_XYTableToPoint"

arcpy.management.XYTableToPoint(in_table=potential_site_TabulateInter, out_feature_class=potential_site_TabulateInter_XYTableToPoint, x_field="AREA", y_field="AREA", coordinate_system="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532925199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAMETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]];-36530900 -28803200 10000;-100000 10000;-100000 10000;0.001;0.001;0.001;IsHighPrecision")

```
if __name__ == '__main__':

    # Global Environment settings

    with arcpy.EnvManager(autoCommit=1000, baUseDetailedAggregation=False,
cellAlignment="DEFAULT",

                          cellSize="MAXOF", cellSizeProjectionMethod="CONVERT_UNITS",
coincidentPoints="MEAN",

                          compression="LZ77", maintainAttachments=True, maintainSpatialIndex=False,

                          matchMultidimensionalVariable=True, nodata="NONE", outputMFlag="Same As
Input",

                          outputZFlag="Same As Input", preserveGlobalIds=False, pyramid="PYRAMIDS -1
NEAREST DEFAULT 75 NO_SKIP NO_SIPS",

                          qualifiedFieldNames=True, randomGenerator="0 ACM599",
rasterStatistics="STATISTICS 1 1",

                          resamplingMethod="NEAREST",
scratchWorkspace="P:\\Final_Nov_24\\Final\\Final.gdb", terrainMemoryUsage=False,

                          tileSize="128 128", tinSaveVersion="CURRENT", transferDomains=False,

                          transferGDBAttributeProperties=False, unionDimension=False,
workspace="P:\\Final_Nov_24\\Final\\Final.gdb"):

        Task1()
```



Model 2: Using the distance accumulation tool to analyze how stormwater energy dissipates from the three gauge stations

Model Script:

Please find the python script that defines the model displayed in Model 2:

```
# -*- coding: utf-8 -*-

"""

Generated by ArcGIS ModelBuilder on : 2024-12-11 12:57:57

"""

import arcpy

from arcpy.ia import *

from arcpy.ia import *

from arcpy.ia import *

from arcpy.sa import *

from arcpy.sa import *
```

```
from arcpy.sa import *

def Task2():  # Task2

    # To allow overwriting outputs change overwriteOutput option to True.

    arcpy.env.overwriteOutput = False



    # Check out any necessary licenses.

    arcpy.CheckOutExtension("spatial")

    arcpy.CheckOutExtension("3D")

    arcpy.CheckOutExtension("ImageAnalyst")

    arcpy.ImportToolbox(r"c:\program files\arcgis\pro\Resources\ArcToolbox\toolboxes\Data
Management Tools.tbx")

    # Model Environment settings

    with arcpy.EnvManager(cellSize="P:\\Final\\PIE_LTER_Data\\flow_cost.img",
extent="246689.79478244 938271.32063448 260649.79478244 949761.32063448
PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GEOGCS[\"GCS_North
_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,
298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532925199433]],PROJEC
TION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.0],PARAMETER[\"F
alse_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAMETER[\"Standard_P
arallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68333333333333],PA
RAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]",
mask="P:\\Final_Nov_24\\Final\\..\\..\\Final\\PIE_LTER_Data\\flow_cost.img",


outputCoordinateSystem="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_20
01\",GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID
[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.01
74532925199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",
200000.0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],
PARAMETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\
",42.68333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]",
scratchWorkspace="P:\\Final_Nov_24\\Final\\..\\..\\Final_Nov_24\\Final",
workspace="P:\\Final_Nov_24\\Final\\Final.gdb"):

        Guage_Stations_csv = "Guage Stations.csv"
```

```
    flow_cost_img = arcpy.Raster("flow_cost.img")

    flow_cost_img_2_ = arcpy.Raster("flow_cost.img")

    flow_cost_img_3_ = arcpy.Raster("flow_cost.img")

    potential_sites_XYTableToPoint_potential_sites_XYTableToPoint =
"potential_sites_XYTableToPoint:potential_sites_XYTableToPoint"

    potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_3_ =
"potential_sites_XYTableToPoint:potential_sites_XYTableToPoint"

    potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_4_ =
"potential_sites_XYTableToPoint:potential_sites_XYTableToPoint"

    Final_gdb = "P:\\Final_Nov_24\\Final\\Final.gdb"

    potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_5_ =
"potential_sites_XYTableToPoint:potential_sites_XYTableToPoint"


    # Process: XY Table To Point (XY Table To Point) (management)

    GuageStations = "P:\\Final_Nov_24\\Final\\Final.gdb\\GuageStations"

    arcpy.management.XYTableToPoint(in_table=Guage_Stations_csv,
out_feature_class=GuageStations, x_field="UTM_X", y_field="UTM_Y",
coordinate_system="PROJCS[\"NAD_1983_UTM_Zone_19N\",GEOGCS[\"GCS_North_American
_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,298.257222
101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532925199433]],PROJECTION[\"Tran
sverse_Mercator\"],PARAMETER[\"False_Easting\",500000.0],PARAMETER[\"False_Northing\",0
.0],PARAMETER[\"Central_Meridian\",-69.0],PARAMETER[\"Scale_Factor\",0.9996],PARAMETER
[\"Latitude_Of_Origin\",0.0],UNIT[\"Meter\",1.0]];-5120900 -9998100 10000;-100000
10000;-100000 10000;0.001;0.001;0.001;IsHighPrecision")


    # Process: Select Layer By Attribute (2) (Select Layer By Attribute) (management)

    Guage_two, Count_2_ =
arcpy.management.SelectLayerByAttribute(in_layer_or_view=GuageStations,
where_clause="Gauge_Number = 25525")


    # Process: Select Layer By Attribute (3) (Select Layer By Attribute) (management)
```

```
    Guage_three, Count_3_ =
arcpy.management.SelectLayerByAttribute(in_layer_or_view=GuageStations,
where_clause="Gauge_Number = 25736")


    # Process: Select Layer By Attribute (Select Layer By Attribute) (management)

    Guage_one, Count =
arcpy.management.SelectLayerByAttribute(in_layer_or_view=GuageStations,
where_clause="Gauge_Number = 27708")


    # Process: Distance Accumulation (Distance Accumulation) (sa)

    Distance_Guage1 = "P:\\Final_Nov_24\\Final\\Distanc_GuageSt1.tif"

    Distance_Accumulation = Distance_Guage1

    Output_Back_Direction_Raster = ""

    Output_Source_Direction_Raster = ""

    Output_Source_Location_Raster = ""

    Distance_Guage1 = arcpy.sa.DistanceAccumulation(Guage_one, "", "", flow_cost_img, "",
"BINARY 1 -30 30", "", "BINARY 1 45", Output_Back_Direction_Raster,
Output_Source_Direction_Raster, Output_Source_Location_Raster, "", "", "", "", "PLANAR")

    Distance_Guage1.save(Distance_Accumulation)

    # Process: Distance Accumulation (2) (Distance Accumulation) (sa)

    Distance_Guage2 = "P:\\Final_Nov_24\\Final\\Distanc_GuageSt2.tif"

    Distance_Accumulation_2_ = Distance_Guage2

    Output_Back_Direction_Raster_2_ = ""

    Output_Source_Direction_Raster_2_ = ""

    Output_Source_Location_Raster_2_ = ""

    Distance_Guage2 = arcpy.sa.DistanceAccumulation(Guage_two, "", "", flow_cost_img_2_,
"", "BINARY 1 -30 30", "", "BINARY 1 45", Output_Back_Direction_Raster_2_,
```

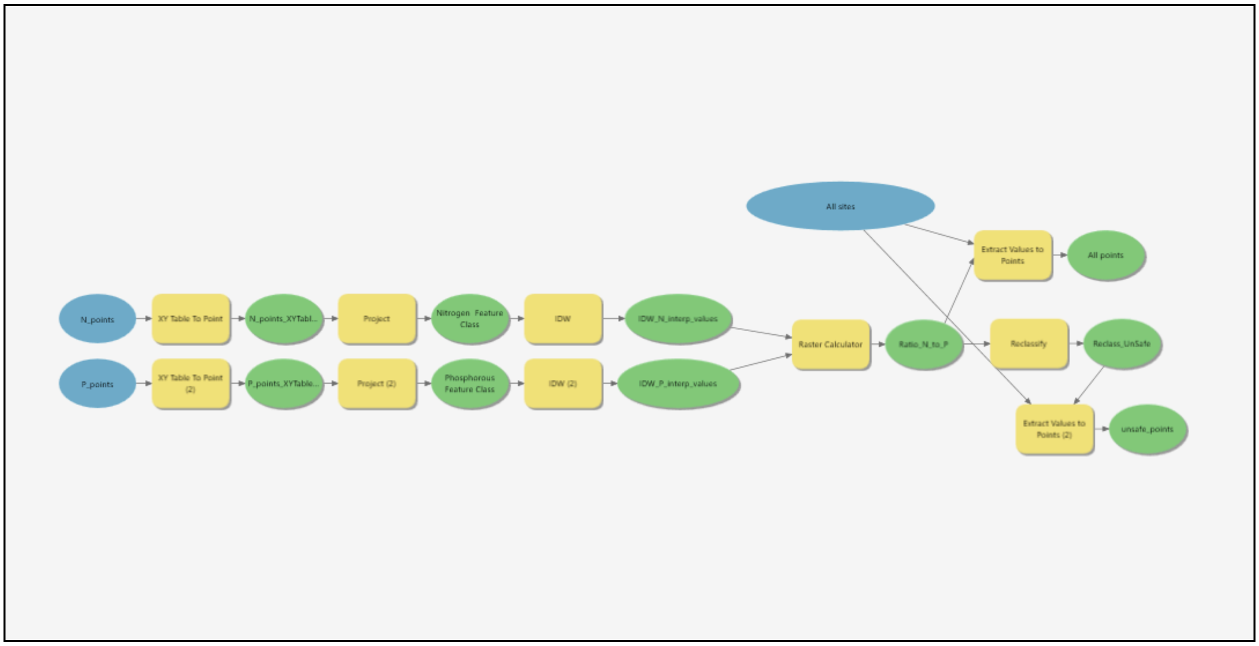
```
Output_Source_Direction_Raster_2_, Output_Source_Location_Raster_2_, "", "", "", "",
"PLANAR")

        Distance_Guage2.save(Distance_Accumulation_2_)

        # Process: Distance Accumulation (3) (Distance Accumulation) (sa)

        Distance_Guage3 = "P:\\Final_Nov_24\\Final\\Distanc_GuageSt3.tif"

        Distance_Accumulation_3_ = Distance_Guage3

        Output_Back_Direction_Raster_3_ = ""

        Output_Source_Direction_Raster_3_ = ""

        Output_Source_Location_Raster_3_ = ""

        Distance_Guage3 = arcpy.sa.DistanceAccumulation(Guage_three, "", "", flow_cost_img_3_,
"", "BINARY 1 -30 30", "", "BINARY 1 45", Output_Back_Direction_Raster_3_,
Output_Source_Direction_Raster_3_, Output_Source_Location_Raster_3_, "", "", "", "",
"PLANAR")

        Distance_Guage3.save(Distance_Accumulation_3_)

        # Process: Reclassify (4) (Reclassify) (3d)

        Reclass_Safe_points_tif = "P:\\Final_Nov_24\\Final\\Reclass_Safe_points.tif"

        arcpy.ddd.Reclassify(in_raster=Distance_Guage1, reclass_field="VALUE", remap="0 748
NODATA;748.010000 5792.838379 1", out_raster=Reclass_Safe_points_tif)

        Reclass_Safe_points_tif = arcpy.Raster(Reclass_Safe_points_tif)



        # Process: Extract Values to Points (Extract Values to Points) (sa)

        Guage1_Safe_points_shp = "P:\\Final_Nov_24\\Final\\Guage1_Safe_points.shp"


arcpy.sa.ExtractValuesToPoints(potential_sites_XYTableToPoint_potential_sites_XYTableToPoint,
Reclass_Safe_points_tif, Guage1_Safe_points_shp, "NONE", "VALUE_ONLY")



        # Process: Reclassify (5) (Reclassify) (3d)
```

```
        Reclass_safepoints2_tif = "P:\\Final_Nov_24\\Final\\Reclass_safepoints2.tif"

        arcpy.ddd.Reclassify(in_raster=Distance_Guage2, reclass_field="VALUE", remap="0 1250 NODATA;1250 6605.396973 1", out_raster=Reclass_safepoints2_tif)

        Reclass_safepoints2_tif = arcpy.Raster(Reclass_safepoints2_tif)


        # Process: Extract Values to Points (2) (Extract Values to Points) (sa)

        Guage_2_safe_points_shp = "P:\\Final_Nov_24\\Final\\Guage_2_safe_points.shp"

arcpy.sa.ExtractValuesToPoints(potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_3_, Reclass_safepoints2_tif, Guage_2_safe_points_shp, "NONE", "VALUE_ONLY")


        # Process: Reclassify (3) (Reclassify) (3d)

        Storm_guage3_tif = "P:\\Final_Nov_24\\Final\\Reclass_Distanc3.tif"

        arcpy.ddd.Reclassify(in_raster=Distance_Guage3, reclass_field="VALUE", remap="0 450 1;450 7575.333008 NODATA", out_raster=Storm_guage3_tif)

        Storm_guage3_tif = arcpy.Raster(Storm_guage3_tif)


        # Process: Raster Calculator (3) (Raster Calculator) (ia)

        Storm_surge3_tif = "P:\\Final_Nov_24\\Final\\Storm_surge3.tif"

        Raster_Calculator_3_ = Storm_surge3_tif

        Storm_surge3_tif =  Storm_guage3_tif * Distance_Guage3

        Storm_surge3_tif.save(Raster_Calculator_3_)


        # Process: Reclassify (6) (Reclassify) (3d)

        Reclass_safepoints3_tif = "P:\\Final_Nov_24\\Final\\Reclass_safepoints3.tif"
```

```
    arcpy.ddd.Reclassify(in_raster=Distance_Guage3, reclass_field="VALUE", remap="0 450 NODATA;450 7575.333008 1", out_raster=Reclass_safepoints3_tif)

    Reclass_safepoints3_tif = arcpy.Raster(Reclass_safepoints3_tif)


    # Process: Extract Values to Points (3) (Extract Values to Points) (sa)

    Guage3_safe_points_shp = "P:\\Final_Nov_24\\Final\\Guage3_safe_points.shp"

arcpy.sa.ExtractValuesToPoints(potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_4_, Reclass_safepoints3_tif, Guage3_safe_points_shp, "NONE", "VALUE_ONLY")


    # Process: Reclassify (Reclassify) (3d)

    Storm_guage1_tif = "P:\\Final_Nov_24\\Final\\Storm_guage1.tif"

    arcpy.ddd.Reclassify(in_raster=Distance_Guage1, reclass_field="VALUE", remap="0 748 1;748 5792.838379 NODATA", out_raster=Storm_guage1_tif)

    Storm_guage1_tif = arcpy.Raster(Storm_guage1_tif)


    # Process: Raster Calculator (Raster Calculator) (ia)

    Storm_surge1_tif = "P:\\Final_Nov_24\\Final\\Storm_surge1.tif"

    Raster_Calculator = Storm_surge1_tif

    Storm_surge1_tif =  Storm_guage1_tif *  Distance_Guage1

    Storm_surge1_tif.save(Raster_Calculator)


    # Process: Reclassify (2) (Reclassify) (3d)

    Stom_guage2_tif = "P:\\Final_Nov_24\\Final\\Reclass_Distanc2.tif"

    arcpy.ddd.Reclassify(in_raster=Distance_Guage2, reclass_field="VALUE", remap="0 1250 1;1250 6605.396973 NODATA", out_raster=Stom_guage2_tif)
```

```
Stom_guage2_tif = arcpy.Raster(Stom_guage2_tif)


# Process: Raster Calculator (2) (Raster Calculator) (ia)

Storm_Surge2_tif = "P:\\Final_Nov_24\\Final\\Raster2.tif"

Raster_Calculator_2_ = Storm_Surge2_tif

Storm_Surge2_tif =  Distance_Guage2 * Stom_guage2_tif

Storm_Surge2_tif.save(Raster_Calculator_2_)




# Process: Mosaic To New Raster (Mosaic To New Raster) (management)

Combined_storm_surge_extents = arcpy.management.MosaicToNewRaster(input_rasters=[Storm_surge1_tif, Storm_Surge2_tif], output_location=Final_gdb, raster_dataset_name_with_extension="Combined_storm_surge_extents", pixel_type="32_BIT_FLOAT", number_of_bands=1, mosaic_method="MINIMUM")[0]

Combined_storm_surge_extents = arcpy.Raster(Combined_storm_surge_extents)


# Process: Mosaic To New Raster (2) (Mosaic To New Raster) (management)

Combined_safe_points = arcpy.management.MosaicToNewRaster(input_rasters=[Reclass_Safe_points_tif, Reclass_safepoints2_tif, Reclass_safepoints3_tif], output_location=Final_gdb, raster_dataset_name_with_extension="Combined_safe_points", number_of_bands=Count)[0]

Combined_safe_points = arcpy.Raster(Combined_safe_points)


# Process: Extract Values to Points (4) (Extract Values to Points) (sa)

safe_points_final_shp = "P:\\Final_Nov_24\\Final\\safe_points_final.shp"
```

```
arcpy.sa.ExtractValuesToPoints(potential_sites_XYTableToPoint_potential_sites_XYTableToPoint
_5_, Combined_safe_points, safe_points_final_shp, "NONE", "VALUE_ONLY")



if __name__ == '__main__':

   # Global Environment settings

   with arcpy.EnvManager(autoCommit=1000, baUseDetailedAggregation=False,
cellAlignment="DEFAULT",

                       cellSizeProjectionMethod="CONVERT_UNITS", coincidentPoints="MEAN",
compression="LZ77",

                       maintainAttachments=True, maintainSpatialIndex=False,
matchMultidimensionalVariable=True,

                       nodata="NONE", outputMFlag="Same As Input", outputZFlag="Same As Input",

                       preserveGlobalIds=False, pyramid="PYRAMIDS -1 NEAREST DEFAULT 75 NO_SKIP
NO_SIPS", qualifiedFieldNames=True,

                       randomGenerator="0 ACM599", rasterStatistics="STATISTICS 1 1",
resamplingMethod="NEAREST",

                       terrainMemoryUsage=False, tileSize="128 128", tinSaveVersion="CURRENT",

                       transferDomains=False, transferGDBAttributeProperties=False,
unionDimension=False):

       Task2()
```



Model 3: Using IDW interpolation to estimate nitrogen and phosphorus concentrations and the N:P ratio for the area based on the interpolated data.

Model Script:

Please find the python script that defines the model displayed in Model 3:

```
# -*- coding: utf-8 -*-

"""

Generated by ArcGIS ModelBuilder on : 2024-12-11 12:59:59

"""

import arcpy

from arcpy.ia import *



def Task3():  # Task3


    # To allow overwriting outputs change overwriteOutput option to True.

    arcpy.env.overwriteOutput = False
```

```
    # Check out any necessary licenses.

    arcpy.CheckOutExtension("3D")

    arcpy.CheckOutExtension("spatial")

    arcpy.CheckOutExtension("ImageAnalyst")


    arcpy.ImportToolbox(r"c:\program files\arcgis\pro\Resources\ArcToolbox\toolboxes\Data
Management Tools.tbx")

    All_sites = "potential_sites_XYTableToPoint:potential_sites_XYTableToPoint"

    N_points = "N_points"

    P_points = "P_points"


    # Process: XY Table To Point (XY Table To Point) (management)

    N_points_XYTableToPoint1 =
"P:\\Final_Nov_24\\Final\\Final.gdb\\N_points_XYTableToPoint1"

    arcpy.management.XYTableToPoint(in_table=N_points,
out_feature_class=N_points_XYTableToPoint1, x_field="Long", y_field="Lat",
coordinate_system="GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_19
83\",SPHEROID[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\
"Degree\",0.0174532925199433]];-400 -400 1000000000;-100000 10000;-100000
10000;8.98315284119521E-09;0.001;0.001;IsHighPrecision")


    # Process: Project (Project) (management)

    Nitrogen_Feature_Class =
"P:\\Final_Nov_24\\Final\\Final.gdb\\N_points_XYTableToPo_Project"

    arcpy.management.Project(in_dataset=N_points_XYTableToPoint1,
out_dataset=Nitrogen_Feature_Class,
out_coor_system="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GE
OGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_
1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.017453292
```

```
5199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.
0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAM
ETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68
333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]")


    # Process: IDW (IDW) (3d)

    IDW_N_interp_values = "P:\\Final_Nov_24\\Final\\Final.gdb\\Idw_N_points1"

    with
arcpy.EnvManager(cellSize="P:\\Final_Nov_24\\PIE_LTER\\PIE_LTER_Data\\flow_cost.img\\flow
_cost", mask="P:\\Final_Nov_24\\PIE_LTER\\PIE_LTER_Data\\flow_cost.img\\flow_cost"):

        arcpy.ddd.Idw(in_point_features=Nitrogen_Feature_Class, z_field="N_Conc",
out_raster=IDW_N_interp_values, cell_size="45.96", power=5)

        IDW_N_interp_values = arcpy.Raster(IDW_N_interp_values)


    # Process: XY Table To Point (2) (XY Table To Point) (management)

    P_points_XYTableToPoint = "P:\\Final_Nov_24\\Final\\Final.gdb\\P_points_XYTableToPoint"

    arcpy.management.XYTableToPoint(in_table=P_points,
out_feature_class=P_points_XYTableToPoint, x_field="Long", y_field="Lat",
coordinate_system="GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_19
83\",SPHEROID[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\
"Degree\",0.0174532925199433]];-400 -400 1000000000;-100000 10000;-100000
10000;8.98315284119521E-09;0.001;0.001;IsHighPrecision")


    # Process: Project (2) (Project) (management)

    Phosphorous_Feature_Class =
"P:\\Final_Nov_24\\Final\\Final.gdb\\P_points_XYTableToPo_Project"

    arcpy.management.Project(in_dataset=P_points_XYTableToPoint,
out_dataset=Phosphorous_Feature_Class,
out_coor_system="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GE
OGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_
1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.017453292
5199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.
```

```
0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAM
ETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68
333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]")


    # Process: IDW (2) (IDW) (3d)

    IDW_P_interp_values = "P:\\Final_Nov_24\\Final\\Final.gdb\\IDW_P_interp_values"

    with
arcpy.EnvManager(cellSize="P:\\Final_Nov_24\\PIE_LTER\\PIE_LTER_Data\\flow_cost.img",
mask="P:\\Final_Nov_24\\PIE_LTER\\PIE_LTER_Data\\flow_cost.img"):

        arcpy.ddd.Idw(in_point_features=Phosphorous_Feature_Class, z_field="P_conc",
out_raster=IDW_P_interp_values)

        IDW_P_interp_values = arcpy.Raster(IDW_P_interp_values)



    # Process: Raster Calculator (Raster Calculator) (ia)

    Ratio_N_to_P = "P:\\Final_Nov_24\\Final\\Final.gdb\\Ratio_N_to_P"

    Raster_Calculator = Ratio_N_to_P

    Ratio_N_to_P =  IDW_N_interp_values / IDW_P_interp_values

    Ratio_N_to_P.save(Raster_Calculator)

    # Process: Extract Values to Points (Extract Values to Points) (sa)

    All_points = "P:\\Final_Nov_24\\Final\\Final.gdb\\Extract_potenti2"

    arcpy.sa.ExtractValuesToPoints(All_sites, Ratio_N_to_P, All_points, "NONE", "VALUE_ONLY")

    # Process: Reclassify (Reclassify) (3d)

    Reclass_UnSafe = "P:\\Final_Nov_24\\Final\\Final.gdb\\Reclass_UnSafe"

    arcpy.ddd.Reclassify(in_raster=Ratio_N_to_P, reclass_field="VALUE", remap="4.254036
12.999999 NODATA;13 18.999999 1;19 109.276955 NODATA", out_raster=Reclass_UnSafe)

    Reclass_UnSafe = arcpy.Raster(Reclass_UnSafe)
```

```
    # Process: Extract Values to Points (2) (Extract Values to Points) (sa)

    unsafe_points = "P:\\Final_Nov_24\\Final\\Final.gdb\\unsafe_points"

    arcpy.sa.ExtractValuesToPoints(All_sites, Reclass_UnSafe, unsafe_points, "NONE",
"VALUE_ONLY")



if __name__ == '__main__':

    # Global Environment settings

    with arcpy.EnvManager(autoCommit=1000, baUseDetailedAggregation=False,
cellAlignment="DEFAULT",

                          cellSize="MAXOF", cellSizeProjectionMethod="CONVERT_UNITS",
coincidentPoints="MEAN",

                          compression="LZ77", extent="246689.79478244 938271.32063448
260649.79478244 949761.32063448
PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GEOGCS[\"GCS_North
_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,2
98.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532925199433]],PROJECTI
ON[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.0],PARAMETER[\"Fal
se_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAMETER[\"Standard_Par
allel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68333333333333],PAR
AMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]", maintainAttachments=True,

                          maintainSpatialIndex=False, matchMultidimensionalVariable=True,
nodata="NONE",


outputCoordinateSystem="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_20
01\",GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[
\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.017
4532925199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",2
00000.0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],P
ARAMETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\"
,42.68333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]]",
outputMFlag="Same As Input", outputZFlag="Same As Input",

                          preserveGlobalIds=False, pyramid="PYRAMIDS -1 NEAREST DEFAULT 75 NO_SKIP
NO_SIPS", qualifiedFieldNames=True,
```

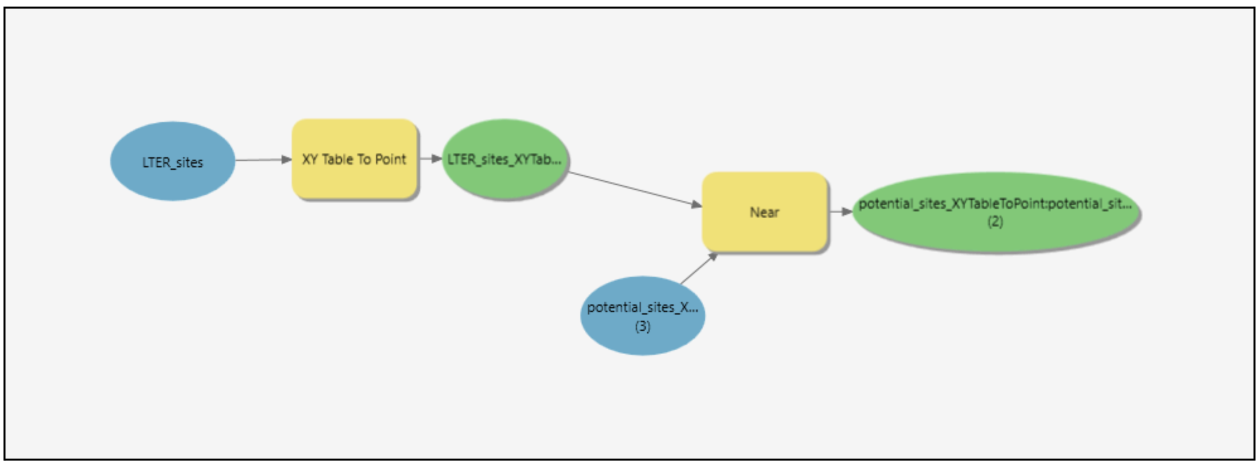
randomGenerator="0 ACM599", rasterStatistics="STATISTICS 1 1", resamplingMethod="NEAREST",

scratchWorkspace="P:\\Final_Nov_24\\Final\\Final.gdb", terrainMemoryUsage=False, tileSize="128 128",

tinSaveVersion="CURRENT", transferDomains=False, transferGDBAttributeProperties=False,

unionDimension=False, workspace="P:\\Final_Nov_24\\Final\\Final.gdb"):

Task3()



Model 4: Using the near tool to identify quahog introduction sites closest to the LTER sites

Model Script:

Please find the python script that defines the model displayed in Model 4:

```
# -*- coding: utf-8 -*-

"""

Generated by ArcGIS ModelBuilder on : 2024-12-08 18:45:02

"""

import arcpy
```

```
def Task4():  # Task4


    # To allow overwriting outputs change overwriteOutput option to True.
    arcpy.env.overwriteOutput = False


    arcpy.ImportToolbox(r"c:\program files\arcgis\pro\Resources\ArcToolbox\toolboxes\Data Management Tools.tbx")
    potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_3_ = "potential_sites_XYTableToPoint:potential_sites_XYTableToPoint"
    LTER_sites = "LTER_sites"


    # Process: XY Table To Point (XY Table To Point) (management)
    LTER_sites_XYTableToPoint = "P:\\Final_Nov_24\\Final\\Final.gdb\\LTER_sites_XYTableToPoint"
    arcpy.management.XYTableToPoint(in_table=LTER_sites, out_feature_class=LTER_sites_XYTableToPoint, x_field="STATE_X", y_field="STATE_Y", coordinate_system="PROJCS[\"NAD_1983_StatePlane_Massachusetts_Mainland_FIPS_2001\",GEOGCS[\"GCS_North_American_1983\",DATUM[\"D_North_American_1983\",SPHEROID[\"GRS_1980\",6378137.0,298.257222101]],PRIMEM[\"Greenwich\",0.0],UNIT[\"Degree\",0.0174532925199433]],PROJECTION[\"Lambert_Conformal_Conic\"],PARAMETER[\"False_Easting\",200000.0],PARAMETER[\"False_Northing\",750000.0],PARAMETER[\"Central_Meridian\",-71.5],PARAMETER[\"Standard_Parallel_1\",41.71666666666667],PARAMETER[\"Standard_Parallel_2\",42.68333333333333],PARAMETER[\"Latitude_Of_Origin\",41.0],UNIT[\"Meter\",1.0]];-36530900 -28803200 10000;-100000 10000;-100000 10000;0.001;0.001;0.001;IsHighPrecision")


    # Process: Near (Near) (analysis)
    potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_2_ = arcpy.analysis.Near(in_features=potential_sites_XYTableToPoint_potential_sites_XYTableToPoint_3_, near_features=[LTER_sites_XYTableToPoint], location="LOCATION", distance_unit="Meters")[0]
```

```
if __name__ == '__main__':

    # Global Environment settings

    with arcpy.EnvManager(scratchWorkspace="P:\\Final_Nov_24\\Final\\Final.gdb",
workspace="P:\\Final_Nov_24\\Final\\Final.gdb"):

        Task4()
```